



Lecture 30

Linear Regression

Announcements

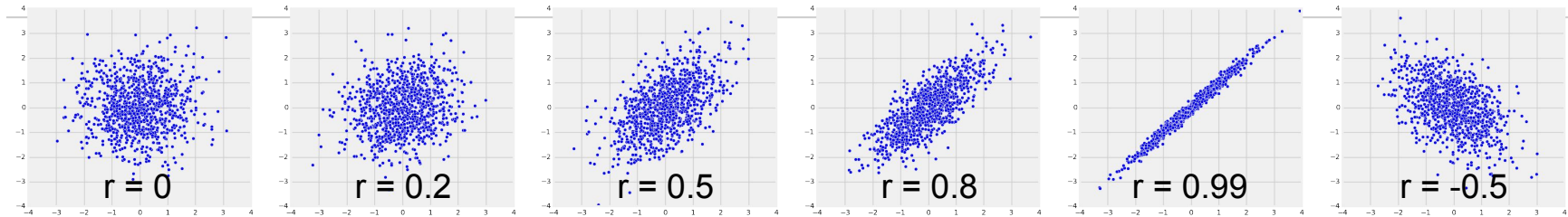
- **Homework 9** is due today at 5pm
 - **Project 2** checkpoint due this Friday
 - Final deadline on April 12th
 - **My office hours:**
 - Changed schedule only for this week ([Ed post](#))
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Weekly Goals

- Monday
 - A measure of linear association
 - **Today**
 - Predicting one numerical variable based on another
 - The regression line
 - Friday
 - The “best” linear predictor
 - The method of least squares
-

Correlation Coefficient

The Correlation Coefficient r



- Measures ***linear*** association
- Based on standard units
- $-1 \leq r \leq 1$
 - $r = 1$: scatter is perfect straight line sloping up
 - $r = -1$: scatter is perfect straight line sloping down
- $r = 0$: No linear association; *uncorrelated*

Definition of r

Correlation Coefficient (r) =

average of	product of	x in standard units	and	y in standard units
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Measures how clustered the scatter is around a straight line

Care in Interpretation

Discussion question

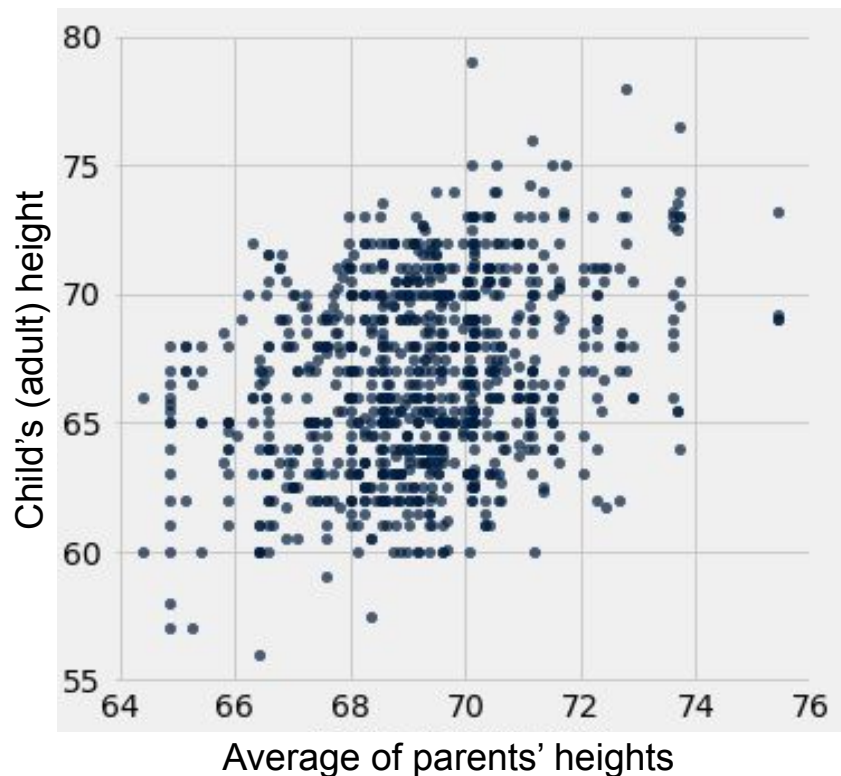
True or False?

If the correlation of x and y is close to 0, then knowing one cannot help us predict the other.

Hint: Go back to the definition correlation coefficient

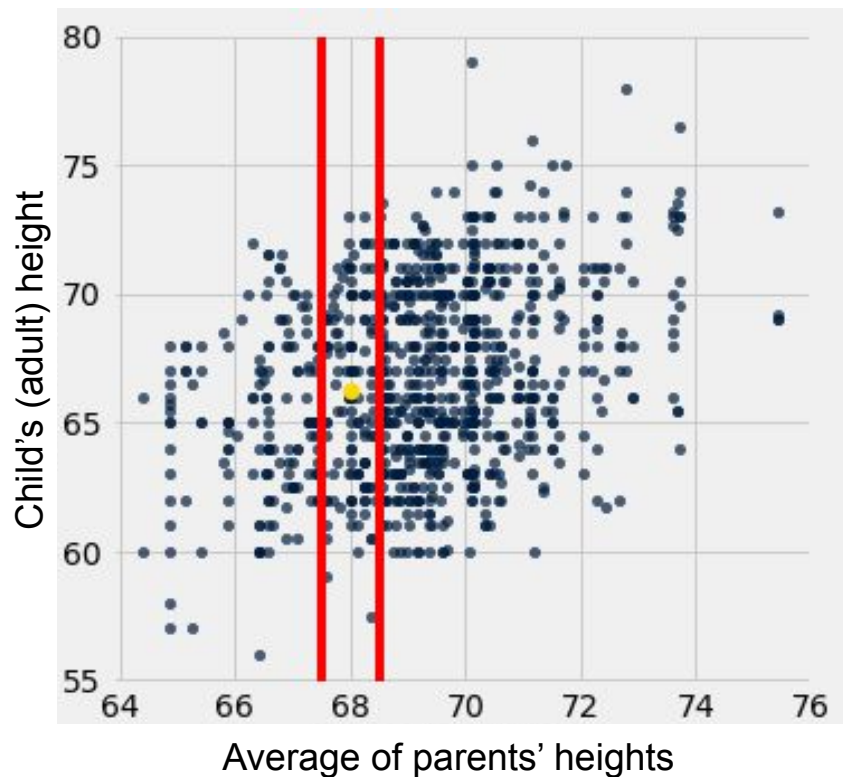
Prediction

Predicting Heights

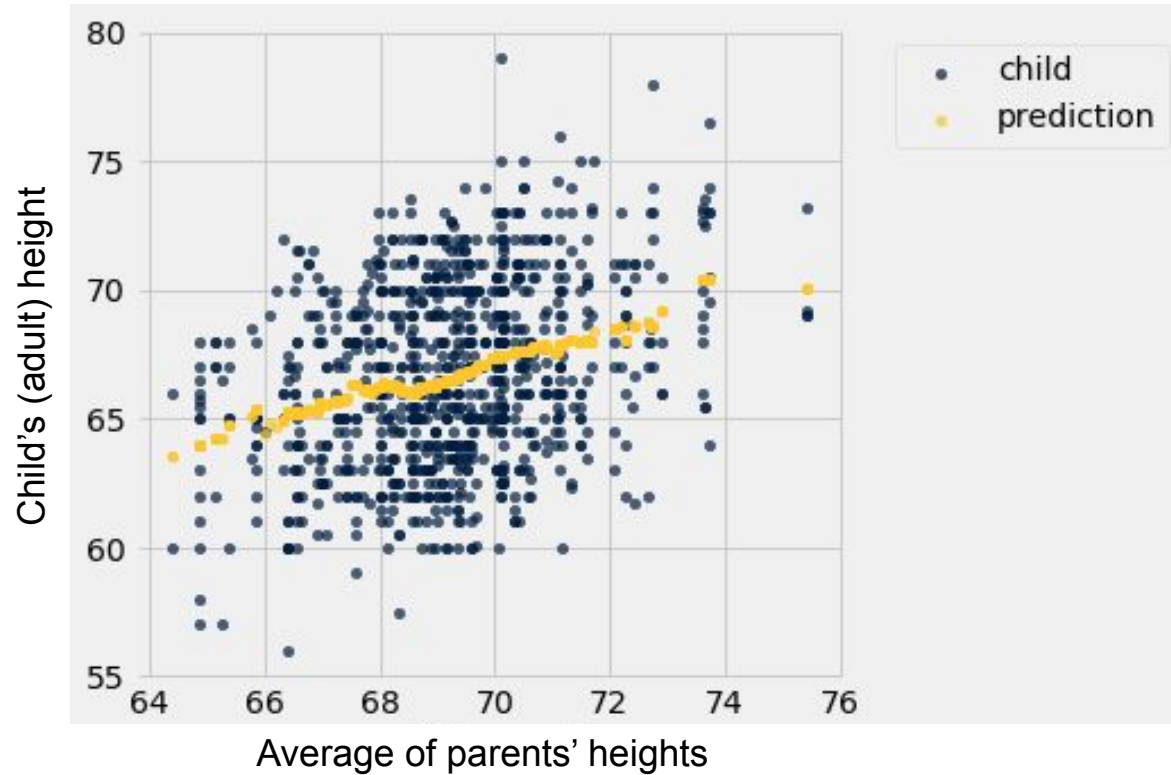


- Oval shaped
- Moderate positive correlation
- How can we predict child height from the parents' average height?

Approach to Prediction



Predicted Heights



Nearest Neighbor Regression

A method for prediction:

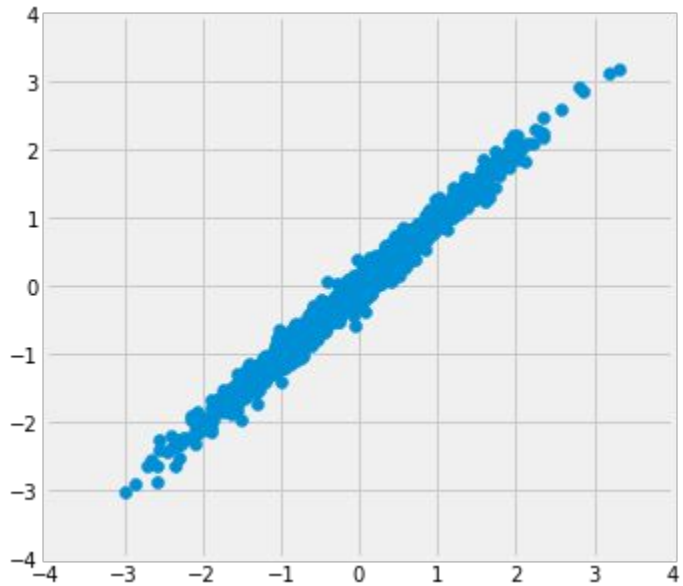
- Group each x with similar (nearby) x values
- Average the corresponding y values for each group

For each x value, the prediction is the average of the y values in its nearby group.

The graph of these predictions is the “graph of averages”.

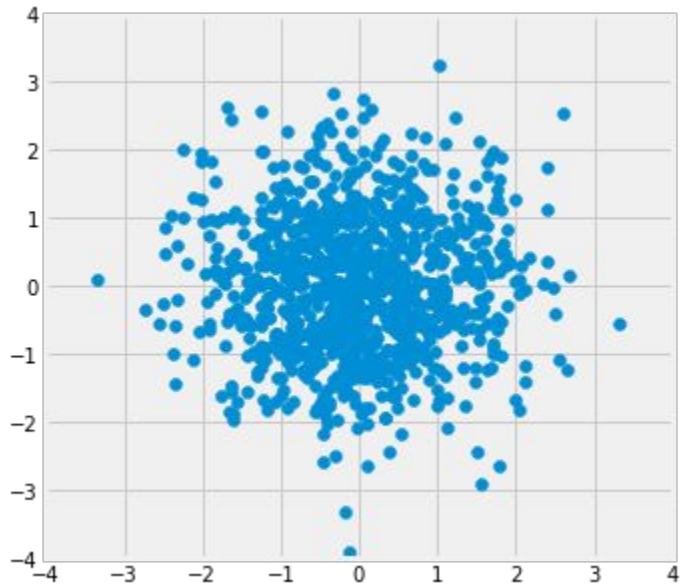
If the association between x and y is linear, then points in the graph of averages tend to fall on a line.

Where is the prediction line?



$$r = 0.99$$

Where is the prediction line?



$r = 0.0$

(Demo)

Linear Regression

Linear Regression

A statement about x and y pairs

- **Measured in *standard units* (su)**
- Describing the deviation of x from 0 (the average of x 's)
- And the deviation of the corresponding y from 0 (the average of y 's)

On average, y deviates from 0 less than x deviates from 0

Regression
line

$$(\text{estimated } y)_{su} = r \times x_{su}$$

correlation

Not true for all points — a statement about averages

Regression Line Equation

In original units, the regression line has this equation:

$$\frac{\text{estimate of } y - \text{average of } y}{\text{SD of } y} = r \times \frac{\text{the given } x - \text{average of } x}{\text{SD of } x}$$

estimated y in standard units

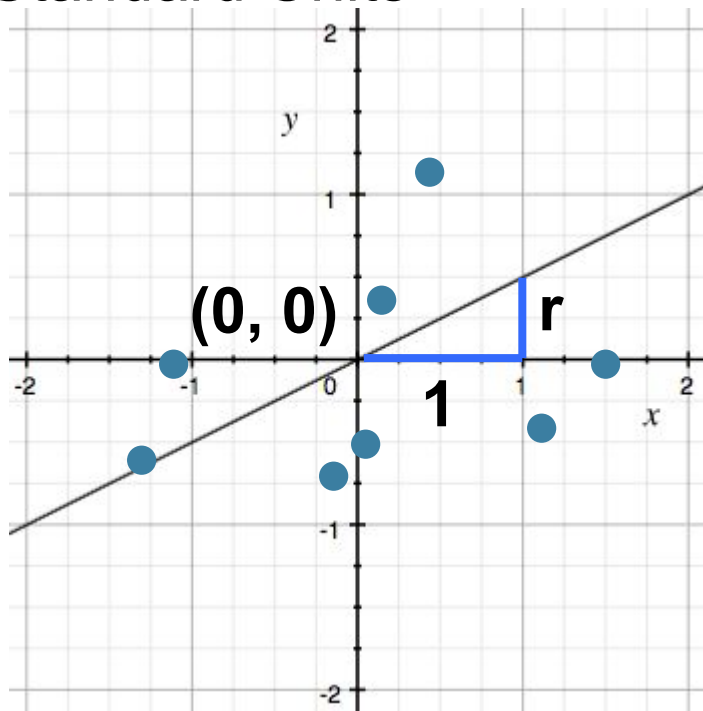
x in standard units

Lines can be expressed by *slope* & *intercept*

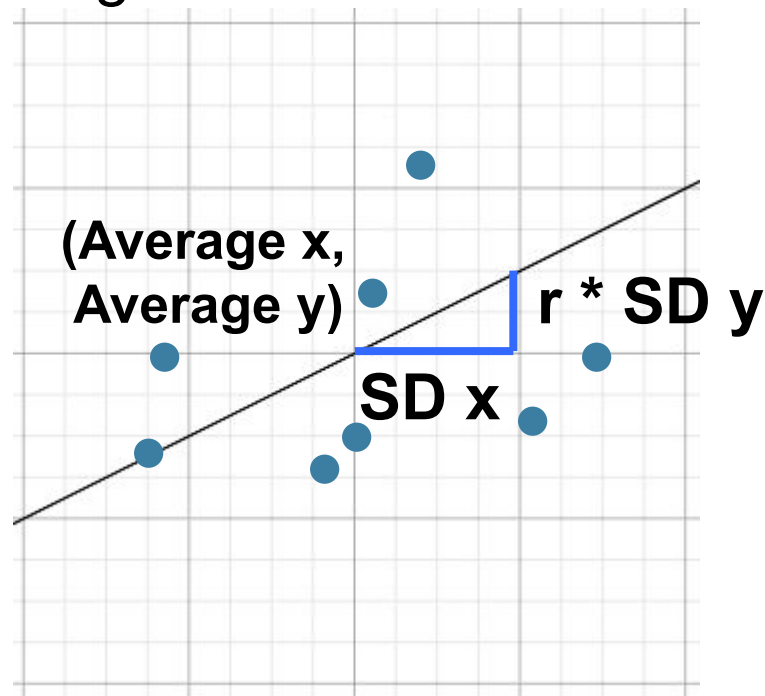
$$y = \text{slope} \times x + \text{intercept}$$

Regression Line

Standard Units



Original Units



Regression Estimate

Goal: Predict y using x

To find the regression estimate of y :

- Convert the given x to standard units
 - Multiply by r
 - That's the regression estimate of y , but:
 - It's in standard units
 - So convert it back to the original units of y
-

Slope and Intercept (in original units)

estimate of y = slope * x + intercept

$$\text{slope of the regression line} = r \cdot \frac{\text{SD of } y}{\text{SD of } x}$$

$$\text{intercept of the regression line} = \text{average of } y - \text{slope} \cdot \text{average of } x$$

(Demo)

Discussion Question

A course has a midterm (average 70; standard deviation 10) and a really hard final (average 50; standard deviation 12)

If the scatter diagram comparing midterm & final scores for students has an oval shape with correlation 0.75, then...

What do you expect the average final score would be for students who scored 90 on the midterm?

How about 60 on the midterm?
